

Visa Core Rules & ISO 8583 Dual-Message Processing Architecture

MTI 0100/0200 Parsing, Bit Map Encoding, STAN/RRN Correlation & Settlement Batching

Standard Setter:	Visa Inc. Worldwide Technical Architecture
Specification ID:	VISA-VIPP-ISO8583
Classification:	Payment Card Network Protocol Specification
Verification:	Integrated in Ariya Sarrafzadeh Production Architecture

1. Dual-Message System (DMS) vs. Single-Message System (SMS)

DMS separates real-time authorization (MTI 0100/0110) from subsequent settlement clearing (BASE II clearing files). SMS combines authorization and clearing in a single real-time capture message (MTI 0200/0210).

2. Primary & Secondary Bit Map Layout

Binary or hex bitmap representing presence of data elements (DE 1 through DE 128). Critical fields include DE 3 (Processing Code), DE 4 (Amount), DE 11 (STAN), DE 37 (RRN), and DE 48 (Private Data).

3. STAN & RRN Transaction Correlation

System Trace Audit Numbers (STAN, 6 digits) and Retrieval Reference Numbers (RRN, 12 digits) guarantee transaction idempotency and correlation across merchant, PSP, switch, and issuing bank logs.

4. Reversal Protocols (MTI 0400/0420)

When socket timeouts or downstream acquirer dropouts occur, automated reversal telegrams trigger immediate debit rollback to eliminate orphaned merchant ledger entries.

Systems Verification Notice: This authoritative specification establishes the invariant compliance baseline for production fintech switches, settlement reconciliation engines, and double-entry ledger implementations documented across the portfolio.